Algebra: Algebraic Fractions

Questions 1–5

Question 1. Simplify the following as far as possible:

a) $\frac{(x+4)(x+3)}{x+3}$

b) $\frac{x^2+11x+30}{x+5}$

c) $\frac{x^2+5x}{x^2+7x+10}$

d) $\frac{x^2+2x-35}{x^2+4x-21}$

e) $\frac{2x^2+x-1}{2x^2+15x-8}$

f) $\frac{3x^2+7x+4}{9x^2-16}$

g) $\frac{2x^2+9x+4}{4x^3-8x^2-5x}$

h) $\frac{2x^2+17x+30}{2x^3+11x^2+15x}$

Question 2. Write as a single fully simplified fraction:

a) $\frac{1}{x}+\frac{3}{2x}$

b) $\frac{1}{x^2}+\frac{1}{x}+\frac{3}{5}$

c) $\frac{1}{x}+\frac{1}{x+1}$

d) $\frac{1}{x+2}+\frac{2}{x+4}$

e) $\frac{3}{x+1}-\frac{1}{x-3}$

f) $\frac{5}{x+2}+\frac{7}{2x-1}$

g) $\frac{3}{2(x+5)}+\frac{1}{4(x-3)}$

h) $\frac{x}{x^2-4}+\frac{5}{x+2}$

Question 3. Simplify and fully factorise the following:

a) $\frac{x^2-5x+4}{x^2-x} \times \frac{2x+3}{5x-20}$

b) $\frac{x-3}{3x^2-6x} \times \frac{3x^2-12}{x^2-9}$

c) $\frac{16x^2-4}{3x-1} \times \frac{3x^2+14x-5}{4x+2}$

d) $\frac{x+2}{x+3} \div \frac{x^2+3x+2}{2x^2-2x-12}$

Question 4. Fully simplify the following:

a) $\frac{\frac{1}{x}}{\frac{1}{x+1}}$

b) $\frac{\frac{1}{x-2}}{\frac{1}{2x}}$

c) $\frac{\frac{1}{x+3}+\frac{1}{x-5}}{\frac{x-5}{x+3}}$

d) $\frac{\frac{1}{x+4}+\frac{1}{x-1}}{\frac{x+2}{x-1}}$

e) $\frac{(3x+1)(5x-1)+4(5x-1)}{\frac{5x-1}{3x+1}}$

Question 5. Prove each of the following:

a) $\frac{3}{x+2} + \frac{x-2}{(x+2)(x+3)} \equiv \frac{4x+7}{(x+2)(x+3)}$

b) $\frac{2x^2-x-1}{x^2-1} - \frac{1}{x^2+x} \equiv \frac{2x-1}{x}$

c) $1 - \frac{1}{x-3} + \frac{2}{x^2-9} \equiv \frac{x^2-x-10}{(x-3)(x+3)}$

d) $\frac{9-4x}{2(x-1)} + \frac{1}{x+3} + 2 \equiv \frac{7x+13}{2(x-1)(x+3)}$